

MATH 211 Analysis III — Problem Session 1

Problem 1

Let $x, y, z \in \mathbb{R}^n$.

- a) If $\|x - z\| < 4$ and $\|y - z\| < 2$, prove that $\|x - y\| < 6$.
- b) If $\|x\| < 3$, $\|y\| < 4$, $\|z\| < 2$, prove that $|x \cdot y - x \cdot z| < 18$.
- c) If $\|x - y\| < 7$ and $\|z\| < 2$, prove that $|x \cdot (y - z) - y \cdot (x - z)| < 14$.

Solution.

a)

$$\begin{aligned}\|x - y\| &\leq \|x - y + z - z\| \\ &\leq \|x - z\| + \|z - y\| \quad (\text{triangle inequality}) \\ &= \|x - z\| + \|y - z\| \\ &< 4 + 2 \\ &= 6\end{aligned}$$

b)

$$\begin{aligned}|x \cdot y - x \cdot z| &\leq |x \cdot y| + |x \cdot z| \\ &\leq \|x\|\|y\| + \|x\|\|z\| \\ &< 3 \cdot 4 + 3 \cdot 2 \\ &= 18\end{aligned}$$

c)

$$\begin{aligned}|x \cdot (y - z) - y \cdot (x - z)| &= |(x \cdot y) - (x \cdot z) - (y \cdot x) - (y \cdot (-z))| \\ &= |(x \cdot y) - (x \cdot z) - (x \cdot y) + (y \cdot z)| \\ &= |(y \cdot z) - (x \cdot z)| \leq |(y - x) \cdot z| \\ &\leq \|y - x\| \cdot \|z\| \\ &< 7 \cdot 2 \\ &= 14\end{aligned}$$

Problem 2

In the below exercises, find equations of the planes satisfying the given conditions.

- a) Passing through $(1, 2, 3)$ and parallel to the plane $3x + y - 2z = 15$.
- b) Passing through the three points $(1, 1, 0)$, $(2, 0, 2)$ and $(0, 3, 3)$.
- c) Passing through the line of intersection of the planes $2x + 3y - z = 0$ and $x - 4y + 2z = -5$ and passing through the point $(-2, 0, -1)$.

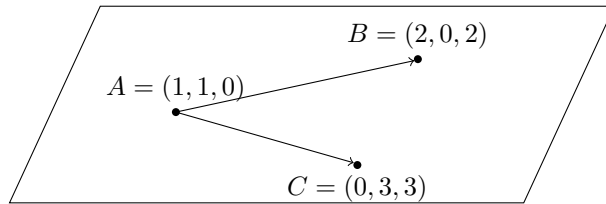
Solution.

a) Let A be the plane passing through $(1, 2, 3)$ and parallel to the plane defined by the equation $3x + y - 2z = 15$. Then a normal of A is the vector $\langle 3, 1, -2 \rangle$.

Equation of A is $3x + y - 2z + d = 0$. For $x = 1, y = 2, z = 3$ we get $d = +1$. It follows that the equation of A is

$$3x + y - 2z + 1 = 0.$$

b)



$$\overrightarrow{AC} = \langle 0, 3, 3 \rangle - \langle 1, 1, 0 \rangle = \langle -1, 2, 3 \rangle.$$

$$\overrightarrow{AB} = \langle 2, 0, 2 \rangle - \langle 1, 1, 0 \rangle = \langle 1, -1, 2 \rangle.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 2 \\ -1 & 2 & 3 \end{vmatrix} = -7\mathbf{i} - 5\mathbf{j} + \mathbf{k} = \langle -7, -5, 1 \rangle$$

Then $n = \langle -7, -5, 1 \rangle$ is normal to the plane passing through the points A, B and C .

$$-7x - 5y + z + d = 0 \quad \text{for} \quad (x, y, z) = (1, 1, 0)$$

$$-7 - 5 + 0 + d = 0 \Rightarrow d = 12.$$

Equation of the plane is $-7x - 5y + z + 12 = 0$.

c) All the planes that contain the intersection have the form

$$\mu(2x + 3y - z) + \lambda(x - 4y + 2z + 5) = 0. \quad \textbf{Why?}$$

Choosing $\mu = 1$ and plugging $(x, y, z) = (-2, 0, -1)$ we get

$$(-4 + 1) + \lambda(-2 - 2 + 5) = 0$$

$$-3 + \lambda(+1) = 0 \Rightarrow \lambda = 3.$$

Equation of the desired plane is

$$2x + 3y - z + 3x - 12y + 6z + 15 = 0$$

$$5x - 9y + 5z = -15.$$

Problem 3

Find an equation of the line passing through $(2, -1, -1)$ and parallel to each of the two planes $x + y = 0$ and $x - y + 2z = 0$.

Solution. Normal n_1 of the plane $x + y = 0$ is $\langle 1, 1, 0 \rangle$ and normal n_2 of the plane $x - y + 2z = 0$ is $\langle 1, -1, 2 \rangle$.

$\langle 1, 1, 0 \rangle \times \langle 1, -1, 2 \rangle$ is parallel to these planes. Then,

$$\langle 1, 1, 0 \rangle \times \langle 1, -1, 2 \rangle = \langle 2, -2, -2 \rangle.$$

So, the direction vector of the line is $\langle 1, -1, -1 \rangle$. Then the vector representation of the line is

$$\langle 2, -1, -1 \rangle + t\langle 1, -1, -1 \rangle = r(t).$$

OR in scalar parametric form:

$$x = 2 + t, \quad y = -1 - t, \quad z = -(1 + t), \quad t \in \mathbb{R}.$$

OR in standard form:

$$x - 2 = -(y + 1) = -(z + 1).$$

Open and Closed Sets

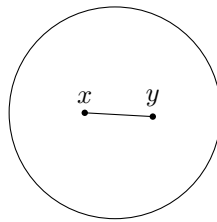
Example.

- a) Every open interval (a, b) on the real line is an open set. In fact if $a < x < b$, choose $\epsilon = \min\{x - a, b - x\}$. Then clearly

$$\text{Br}(x, \epsilon) \subset (a, b)$$

Example.

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$



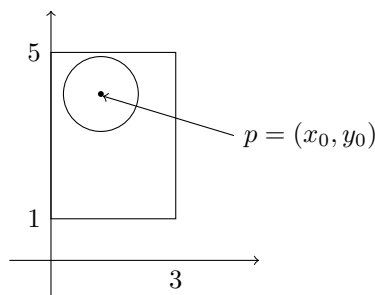
Let x be a point in S^1 . We claim $\text{Br}(x, 1 - \|x\|)$ is a subset of S^1 . Indeed, if $y \in \text{Br}(x, 1 - \|x\|)$ then:

$$\|y\| = \|y - x + x\| \leq \|y - x\| + \|x\| < 1 - \|x\| + \|x\| = 1.$$

The first inequality is by triangle inequality and the second inequality holds since $y \in \text{Br}(x, 1 - \|x\|)$. Thus, S^1 is an open set.

Example.

$$B = \{(x, y) \in \mathbb{R}^2 : 0 < x < 3, 1 < y < 5\}$$



$$B(p, \epsilon) \subseteq B$$

where

$$\epsilon = \min\{x_0 - 0, 3 - x_0, y_0 - 1, 5 - y_0\}.$$

Example.

$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

is not open because, suppose $B(p, \epsilon) \subseteq C$. Then there exists $(a, b) \in B(p, \epsilon)$ which satisfy $\|(a, b)\| > 4$, but this is a contradiction.

Closed Set Examples.

- a) The empty set $\emptyset$ and $\mathbb{R}$ are closed sets (and open sets).
- b) Every closed interval $[a, b]$ on the real line is a closed set.
- c) Any set consisting of a finite number of points is closed.

Exercise. Show that $\mathbb{Q}$ is neither open nor closed set and conclude that irrationals are also not open and closed sets.